1.1 The Brachistochrone Problem

In the study of geometrical optics, Fermat proposed that light always propagates along the path that requires the least time. This principle later became known as **Fermat's Principle**. At the time, however, its profound physical implications had not yet been fully recognized. It was not until Bernoulli posed the brachistochrone problem and Lagrange subsequently solved it using the calculus of variations that the deep connection between the underlying ideas of variational calculus and Fermat's Principle gradually became apparent. We shall begin with the brachistochrone problem as a concrete example to introduce the calculus of variations.

The brachistochrone problem can be stated as follows: given a fixed starting point and a fixed endpoint, and assuming a constant gravitational acceleration, what path should a small object follow so that the time required to travel from the starting point to the endpoint is minimized? The problem is illustrated in Fig.1.1.

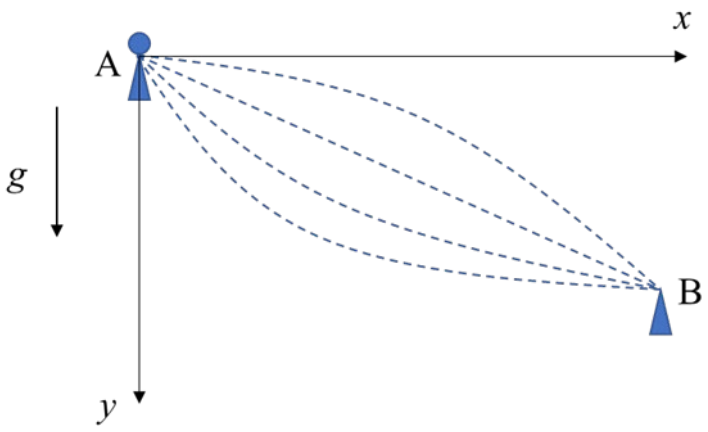


Fig.1.1 the brachistochrone problem

For now, let us assume that such a path of minimum travel time exists, and represent the shape of each possible path as a function. We then introduce a coordinate system similar to the one shown in Fig.1.1, namely,

$$y = f(x) \tag{1.1.1}$$

Considering arbitrary deformations of the curve, we can construct a family of

functions, $\{y_n\} = \{f_1(x), f_2(x), \dots, f_n(x)\}$, where each function passes through the two fixed points A and B. According to the law of conservation of mechanical energy, we have

$$mgy = \frac{1}{2}mv^2 \rightarrow v = \sqrt{2gy} \tag{1.1.2}$$

For an arbitrary infinitesimal element along the trajectory, we decompose it into its components as illustrated in Fig.1.2. The differential element of the path can therefore be expressed as

$$ds = \sqrt{1 + y'^2}dx \tag{1.1.3}$$

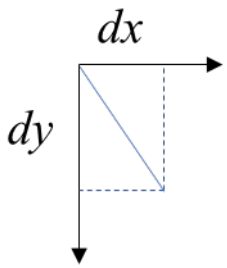


Fig.1.2 the differential element of the path

The total time taken by the object to travel along the path from point A to point B can therefore be expressed as

$$t = \int_A^B \frac{ds}{v} = \int_A^B \frac{\sqrt{1 + y'^2}}{\sqrt{2gy}}dx \tag{1.1.4}$$

If the trajectory is subjected to an infinitesimal variation, such that $\bar{y} = y + \delta y$, the corresponding change in the travel time can be expressed as

$$\begin{aligned}\delta t = \bar{t} - t &= \int_A^B \frac{d\bar{s}}{\bar{v}} - \int_A^B \frac{ds}{v} = \int_A^B \frac{\sqrt{1 + \bar{y}'^2}}{\sqrt{2g\bar{y}}}dx - \int_A^B \frac{\sqrt{1 + y'^2}}{\sqrt{2gy}}dx \\ &= \int_A^B \frac{\sqrt{1 + \bar{y}'^2}}{\sqrt{2g\bar{y}}} - \frac{\sqrt{1 + y'^2}}{\sqrt{2gy}}dx\end{aligned} \tag{1.1.5}$$

We introduce a functional (that is, a function of a function, whose argument is itself a function).

$$F(y, y') = \sqrt{\frac{1 + y'^2}{2gy}} \tag{1.1.6}$$

Substituting Eq. (1.1.6) into Eq. (1.1.5), we obtain

$$\delta t = \int_A^B \left(\frac{\partial F}{\partial y} \delta y + \frac{\partial F}{\partial y'} \delta y' \right) dx \tag{1.1.7}$$

Using the commutativity of variation and differentiation, we obtain

$$\delta y' = \delta \frac{dy}{dx} = \frac{d\delta y}{dx} \tag{1.1.8}$$

Substituting Eq. (1.1.8) into Eq. (1.1.7), we obtain

$$\begin{aligned} \delta t &= \int_A^B \frac{\partial F}{\partial y} \delta y dx + \int_A^B \frac{\partial F}{\partial y'} d\delta y \\ &= \int_A^B \frac{\partial F}{\partial y} \delta y dx + \frac{\partial F}{\partial y'} \delta y \Big|_A^B - \int_A^B \delta y d \frac{\partial F}{\partial y'} \end{aligned} \tag{1.1.9}$$

The above expression is obtained by applying integration by parts. Furthermore, since the two endpoints are fixed, the second term on the far right-hand side of Eq. (1.1.9) vanishes. Therefore, Eq. (1.1.9) can be simplified to

$$\delta t = \int_A^B \left(\frac{\partial F}{\partial y} - \frac{d}{dx} \frac{\partial F}{\partial y'} \right) \delta y dx \tag{1.1.10}$$

Since the variation δy is arbitrary, requiring δt implies that the expression inside the parentheses in Eq. (1.1.10) must vanish identically. That is,

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \frac{\partial F}{\partial y'} = 0 \tag{1.1.11}$$

Substituting Eq. (1.1.6) into Eq. (1.1.11), we obtain

$$\sqrt{\frac{1 + y'^2}{2g}} \cdot \left(-\frac{1}{2} \right) \cdot y^{-\frac{3}{2}} - \frac{d}{dx} \left(\frac{y'}{\sqrt{2gy} \cdot \sqrt{1 + y'^2}} \right) = 0 \tag{1.1.12}$$

After simplifying Eq. (1.1.12), we arrive at

$$2yy'' + y'^2 + 1 = 0 \tag{1.1.13}$$

This is a nonhomogeneous differential equation, which is somewhat difficult to solve directly. However, Eq. (1.1.11) is, strictly speaking, known as the **first form of the Euler–Lagrange equation**. If we differentiate the functional in Eq. (1.1.6), we obtain

$$dF(y, y') = \frac{\partial F}{\partial y} dy + \frac{\partial F}{\partial y'} dy' \tag{1.1.14}$$

Substituting Eq. (1.1.11) into Eq. (1.1.14) gives

$$dF(y,y') = \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)dy + \frac{\partial F}{\partial y'}dy' = d\left(\frac{\partial F}{\partial y'}\right)\cdot y' + \frac{\partial F}{\partial y'}dy' \\ = d\left(y'\cdot\frac{\partial F}{\partial y'}\right) \tag{1.1.15}$$

Integrating both sides of Eq. (1.1.15), we obtain

$$F - y'\cdot\frac{\partial F}{\partial y'} = C \tag{1.1.16}$$

Equation (1.1.16) is also known as the **second form of the Euler–Lagrange equation**. Substituting the expression of the functional into Eq. (1.1.16), we obtain a new differential equation:

$$\sqrt{\frac{1+y'^2}{2gy}} - y'\cdot\frac{y'}{\sqrt{2gy\cdot\left(1+y'^2\right)}} = \frac{1}{\sqrt{2gy\cdot\left(1+y'^2\right)}} = C_1 \tag{1.1.17}$$

Equation (1.1.17) is relatively easier to solve. Expanding it yields

$$x = \int dx = \int\sqrt{\frac{2gC_1^2\cdot y}{1-2gC_1^2\cdot y}}dy + C_2 \tag{1.1.18}$$

Equation (1.1.18) can be evaluated using elementary integration. Solving it and adopting the coordinate convention illustrated in Fig. 1, we obtain $C_2 = 0$, and hence

$$\begin{cases} y = \dfrac{1-cos\theta}{4gC_1^2} \\ x = \dfrac{\theta - sin\theta}{4gC_1^2} \end{cases} \tag{1.1.19}$$

Equation (1.1.19) gives the parametric equations of a **cycloid**, demonstrating that the brachistochrone is a cycloid.